Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Area bounded by $y = \pi \left[\sin^2 \frac{x}{2} + \sqrt{\cos \frac{x}{2}}\right]$ and $y = \left|\sin x - \frac{1}{2}\right|, x = 0, x = \pi$ where [.] represents the greatest integer function is.

(A) $\frac{1}{6}(7\pi + 12(1 - \sqrt{3}))$

(B) $7\pi + 12(1 + \sqrt{3})$

(C) $\frac{\pi}{2} + 1 - \sqrt{3}$

(D) $\frac{\pi}{2} + 1 + \sqrt{3}$

Q.2

Area bounded by the curve $y = \max\{\sin x, \cos x\}$ and x-axis, between the lines $x = \pi/4$ and $x = 2\pi$ is equal to

(A) $\frac{(4\sqrt{2}-1)}{\sqrt{2}}$ sq. units (

(B) $(4\sqrt{2} - 1)$ sq.units

(C) $\frac{(4\sqrt{2}-1)}{2}$ sq. units

(D) None of these

Q.3

Area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$ is

(A) 3

(B) $\frac{8}{3}$

(C) $\frac{7}{3}$

(D) $\frac{10}{3}$

Q.4

Area bounded by the curves $y = |x| - 2$ and $y = 1 - |x - 1|$ is equal to -

(A) 4 sq. units

(B) 6 sq. units

(C) 2 sq. units

(D) 8 sq. units

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Q.5

The area bounded by the curves $y = ex \ln x$ and $y = \frac{\ln x}{ex}$ is

(A) $\frac{e}{4} + \frac{5}{4e}$

(B) $\frac{e}{4} - \frac{5}{4e}$

(C) $\frac{e}{4} - \frac{4}{5e}$

(D) $\frac{e}{4} + \frac{4}{5e}$

Q.6

The area between the curve $y = 2x^4 - x^2$, the x - axis and the ordinates of the two minimum-points on the curve is

(A) $\frac{7}{12}$

(B) $\frac{7}{30}$

(C) $\frac{7}{60}$

(D) $\frac{7}{120}$

Q.7

A rea bounded by the curves $y = \sin x$, tangent drawn to it at $x = 0$ and the line $x = \pi/2$, is equal to

(A) $\frac{\pi^2-4}{2}$ sq. units

(B) $\frac{\pi^2-8}{8}$ sq. units

(C) $\frac{\pi^2-2}{4}$ sq.units

(D) $\frac{\pi^2-2}{2}$ sq. units

Q.8

The area between the curves $y = \sqrt{x}$ and $y = x$ is

(A) $1/3$

(B) $1/6$

(C) $2/3$

(D) 1

Q.9

Consider the curve whose equation is $\mathrm{y} = 1 + 4\mathrm{x} - \mathrm{x}^2$.

(A) A rea between the curve and the x-axis is less than $10\sqrt{5}$ sq. units

(B) A rea between the curve, the y-axis and the line through the origin and that highest point of the curve is $\frac{7}{3}$ sq. units.

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(C) If the line $y = mx$ bisects the area between the curve, $y = 0, x = 0$ and $x = \frac{3}{2}$, then $m = \frac{13}{16}$

(D) The curve is symmetrical about the line $x = 2$.

Q.10

For which of the following values of m, is the area of the region bounded by the curve $y = x - x^2$ and the line $y = x - x^2$ and the line $y = mx$ equals $9/2$?

(A) -4

(B) -2

(C) 2

(D) 4

Q.11

The area of the region bounded between the curves $|y| - |\sin x| \geq 0$ and $x^2 + y^2 - \pi^2 \leq 0$ is $\pi^3 - A$. Find the values greater than A.

(A) 9

(B) 10

(C) 11

(D) 12

Q.12

Passage

Two curves $C_1 \equiv [\mathrm{f(y)}]^{2/3} + [\mathrm{f(x)}]^{1/3} = 0$ and $C_2 \equiv [\mathrm{f(y)}]^{2/3} + [\mathrm{f(x)}]^{2/3} = 12$, satisfying the relation $(\mathrm{x - y})\mathrm{f(x + y)} - (\mathrm{x + y})\mathrm{f(x - y)} = 4\mathrm{xy}\left(\mathrm{x^2 - y^2}\right)$.

Question:

The area bounded by C_1 and C_2 is -

(A) $2\pi - 3\sqrt{3}$ sq. units

(B) $2\pi + \sqrt{3}$ sq. units

(C) $\pi + \sqrt{6}$ sq. units

(D) $2\sqrt{3} - \pi$ sq. units

Q.13

Passage

Two curves $C_1 \equiv [f(y)]^{2/3} + [f(x)]^{1/3} = 0$ and $C_2 \equiv [f(y)]^{2/3} + [f(x)]^{2/3} = 12$, satisfying the relation $(x - y)f(x + y) - (x + y)f(x - y) = 4xy\left(x^2 - y^2\right)$.

Question:

The area bounded by the curve C_2 and $|x| + |y| = \sqrt{12}$ is -

(A) $12\pi - 24$ sq. units

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(B) $6-\sqrt{12}$ sq. units

(C) $2\sqrt{12}-6$ sq. units

(D) None of these

Q.14

Passage

Two curves $C_1 \equiv [f(y)]^{2/3}+[f(x)]^{1/3}=0$ and $C_2 \equiv [f(y)]^{2/3}+[f(x)]^{2/3}=12$, satisfying the relation $(x-y)f(x+y)-(x+y)f(x-y)=4xy\left(x^2-y^2\right)$.

Question:

The area bounded by C_1 and $x+y+2=0$ is -

(A) $5/2$ sq. units

(B) $7/2$ sq. units

(C) $9/2$ sq. units

(D) None of these

Q.15

Passage

Consider the function defined implicity by the equation $\mathrm{y}^2-2\mathrm{ye}^{\sin^{-1}\mathrm{x}}+\mathrm{x}^2-1+[\mathrm{x}]+\mathrm{e}^{2\sin^{-1}\mathrm{x}}=0$, (where $[x]$ denotes the greatest integer function).

Question:

The area of the region bounded by the curve and the line $x=-1$ is -

(A) $\pi+1$ sq. units

(B) $\pi-1$ sq. units

(C) $\frac{\pi}{2}+1$ sq. units

(D) $\frac{\pi}{2}-1$ sq. units

Q.16

Passage

Consider the function defined implicity by the equation $y^2-2ye^{\sin^{-1}x}+x^2-1+[x]+e^{2\sin^{-1}x}=0$, (where $[\mathrm{x}]$ denotes the greatest integer function).

Question:

Line $x=0$ divides the region mention above in two parts. The ratio of area of the left-hand side of line to that of right-hand side of line is -

(A) $1+\pi:\pi$

(B) $2-\pi:\pi$

(C) $1:1$

(D) $\pi+2:\pi$

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Q.17

Passage

Consider the function defined implicity by the equation $y^2 - 2ye^{\sin^{-1}x} + x^2 - 1 + [x] + e^{2\sin^{-1}x} = 0$, (where $[x]$ denotes the greatest integer function).

Question:

The area of the region of curve and the line $x = 0$ and $x = \frac{1}{2}$ is –

(A) $\frac{\sqrt{3}}{4} + \frac{\pi}{6}$ sq. units

(B) $\frac{\sqrt{3}}{2} + \frac{\pi}{6}$ sq. units

(C) $\frac{\sqrt{3}}{4} - \frac{\pi}{6}$ sq. units

(D) $\frac{\sqrt{3}}{2} - \frac{\pi}{6}$ sq. units

Q.18

Match the column -

(A)	Area bounded by the curves $y = \left[\frac{x^2}{64} + 2\right]$ ([.] denotes the greatest integer function), $y = x - 1$ and $x = 0$ above the x-axis is	(p) 4
(B)	The area bounded by $y = \log_e x$, x-axis and the ordinate $x = e$ is given by –	(q) 1
(C)	The area bounded by the curve $y = \sin^{-1}x$ and the lines $x = 0, \lvert y \rvert = \pi/2$ is –	(r) 2
(D)	The total area enclosed by the lines $y = \lvert x \rvert$, $y = 0$ and $\lvert x \rvert = 1$ is –	(s) 3

(A) A-p ; B-q ; C-q ; D-r

(B) A-p ; B-q ; C-r ; D-q

(C) A-p ; B-r ; C-r ; D-q

(D) A-q ; B-s ; C-s ; D-r

Q.19

Consider the collection of all curves of the form $y = a - bx^2$ that passes through the point $(2, 1)$ where a and b are positive real numbers. If the minimum area of the region bounded by $y = a - bx^2$ and the x-axis is $\sqrt{A}$, then find the value of $\frac{A}{8}$.

Q.20

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If the tangent to the curve $y = 1 - x^2$ at $x = \alpha$, where $0 < \alpha < 1$, meets the axes at P and Q. If α varies and the minimum value of the area of the triangle OPQ is k times the area bounded by the axes and the part of the curve $0 < x < 1$, then find $\sqrt{3}k$.

Answers & Solutions

Q.1 Answer:

—

Solution:

$1 \leq \sin^2 \frac{x}{2} + \sqrt{\cos\left(\frac{x}{2}\right)} < 2. \Rightarrow \left[\sin^2 \frac{x}{2} + \sqrt{\cos \frac{x}{2}}\right] = 1$

$\therefore$ required area is

$$= \pi - 2\left(\int_0^{\pi/6} \left(\frac{1}{2} - \sin x\right) dx + \int_{\pi/6}^{\pi/2} \left(\sin x - \frac{1}{2}\right) dx\right)$$

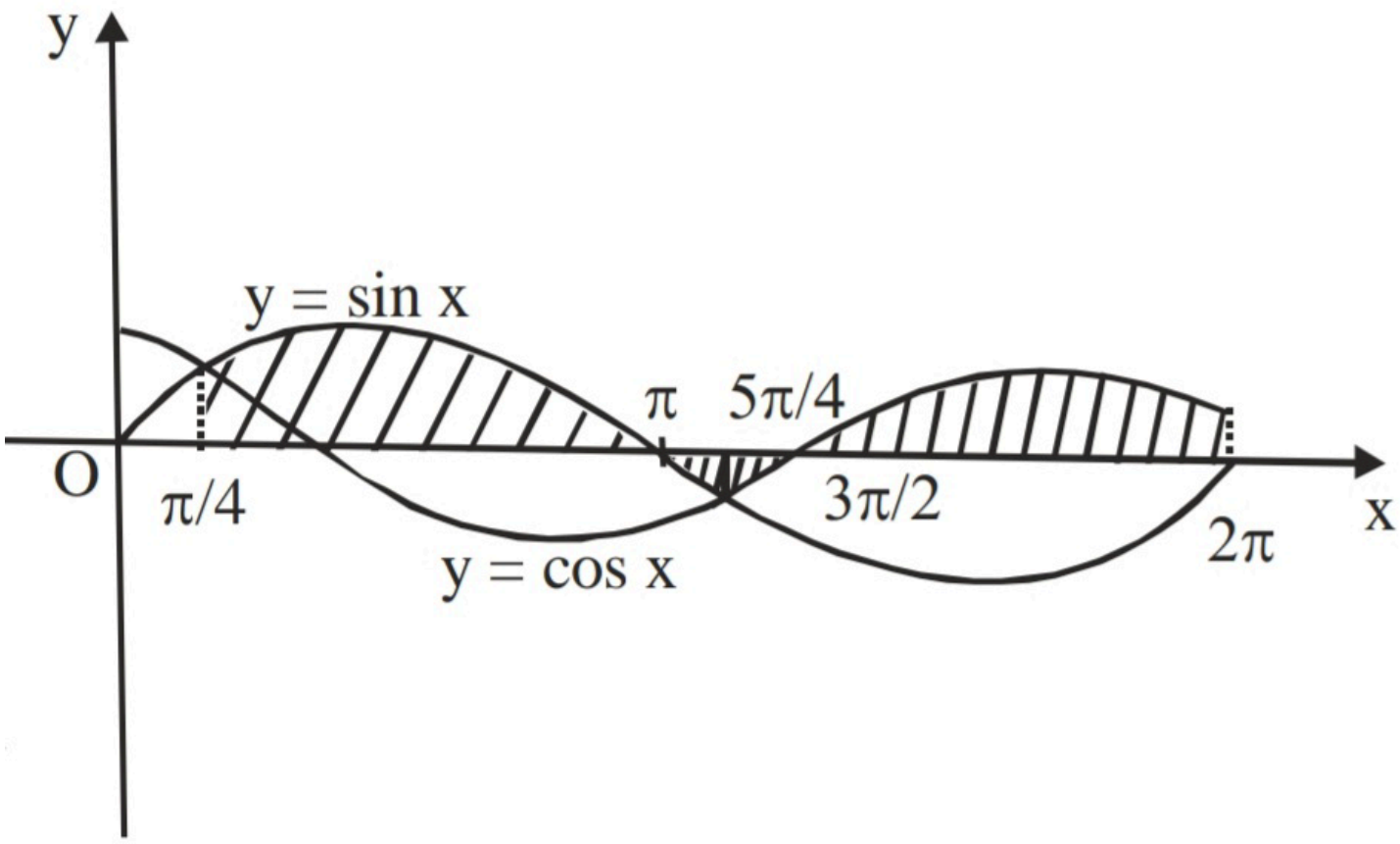
$$= \frac{7\pi}{6} + 2(1 - \sqrt{3})$$

Q.2 Answer:

$\frac{(4\sqrt{2}-1)}{\sqrt{2}}$
sq. units (

Solution:

Bold lines represents the graph of $y = \max\{\sin x, \cos x\}$. Required area,



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$\Delta = \int_{\pi/4}^{\pi} \sin x dx - \int_{\pi}^{5\pi/4} \sin x dx - \int_{5\pi/4}^{3\pi/2} \cos x dx + \int_{3\pi/2}^{2\pi} \cos x dx = \frac{(4\sqrt{2}-1)}{\sqrt{2}}$ **sq.units**

Q.3 Answer:

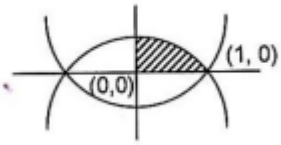
$\frac{8}{3}$

Solution:

$y^2 = (x^2 - 1)^2$

So $y = x^2 - 1$ or $y = 1 - x^2$

So required area $= 4\int_0^1 (1 - x^2)\, dx = 4\left[x - \frac{x^3}{3}\right]_0^1 = \frac{8}{3}$

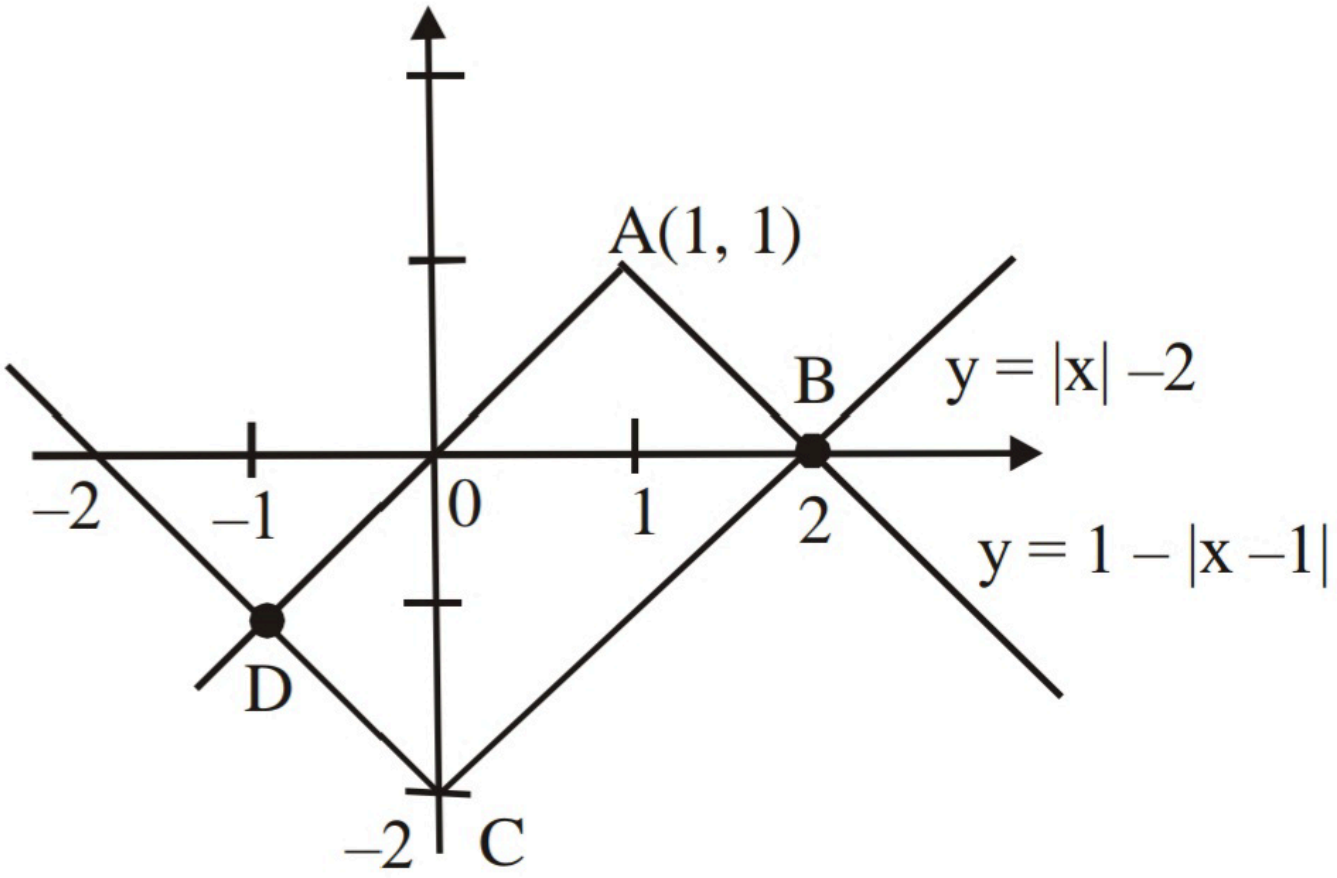


Q.4 Answer:

4 sq. units

Solution:

Bounded figure ABCD **is a rectangle.**



$AB = \sqrt{1+1} = \sqrt{2}, BC = \sqrt{4+4} = 2\sqrt{2}$ **Thus, bounded area** $= (\sqrt{2})(2\sqrt{2}) = 4$ **sq. units.**

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Q.5 Answer:

$\frac{e}{4}-\frac{5}{4e}$

Solution:

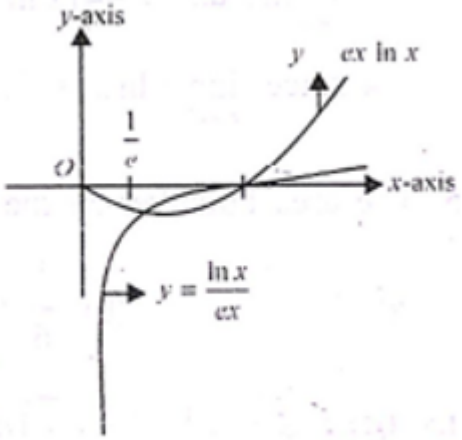
The curves meet at x given by ex $\ln x=\frac{\ln x}{ex}$

$$\Rightarrow x=\frac{1}{e}$$

$$A=\int_{\frac{1}{e}}^{1}\left(\frac{\ln x}{ex}-ex\ln x\right)dx$$

$$=\frac{1}{2e}\Big|_{\frac{1}{e}}^{1}(\ln x)^2-e\left(\frac{x^2}{2}\ln x-\frac{x^2}{4}\right)\Big|_{\frac{1}{e}}^{1}$$

$$=-\frac{1}{2e}+\frac{e}{4}-\frac{3}{4e}=\frac{e}{4}-\frac{5}{4e}$$



Q.6 Answer:

$\frac{7}{120}$

Solution:

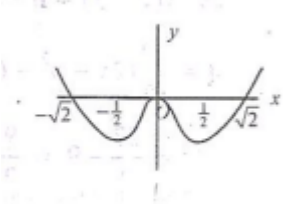
$y=2x^4-x^2, y'=8x^3-2x=2x\left(4x^2-1\right)$

$y'=0\Rightarrow x=0,\pm\frac{1}{2}$

y is maximum at $x=0$

and minimum at $x=\pm\frac{1}{2}$

$$A=\int_{-\frac{1}{2}}^{\frac{1}{2}}\left(x^2-2x^4\right)dx=2\int_0^{\frac{1}{2}}\left(x^2-2x^4\right)dx=2\left[\frac{x^3}{3}-\frac{2}{5}x^5\right]_0^{\frac{1}{2}}=2\left[\frac{1}{24}-\frac{1}{80}\right]=\frac{7}{120}$$



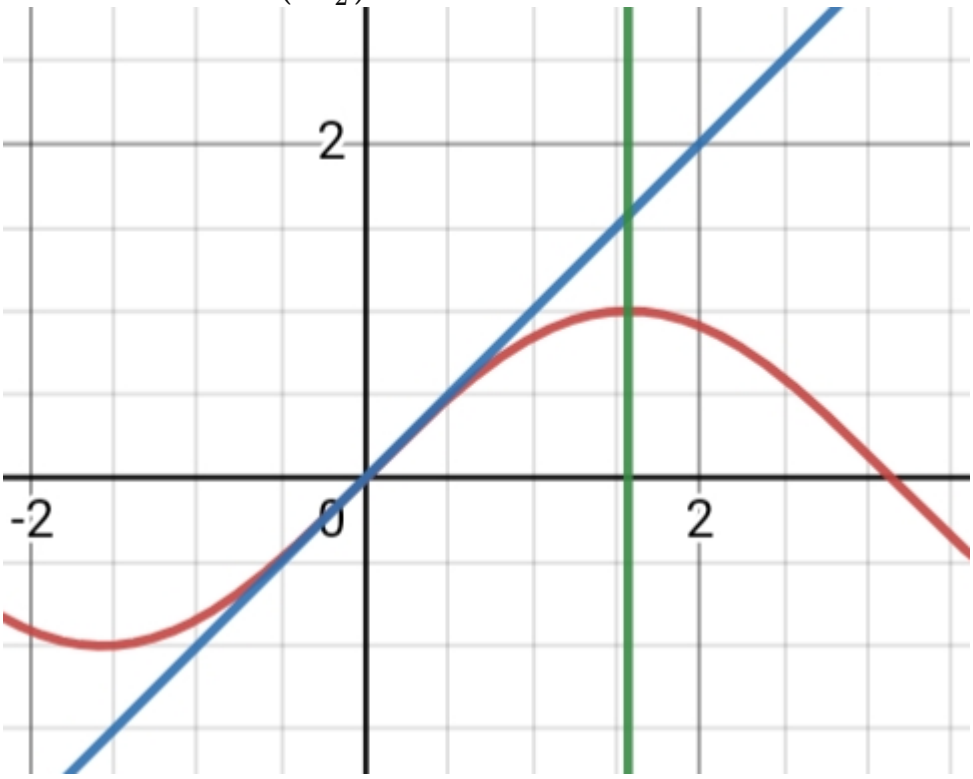
Q.7 Answer:

$\frac{\pi^2-8}{8}$ sq. units

Solution:

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The tangent drawn to $y = \sin x$ at $x = 0$ is the line $y = x$. Clearly the line $y = x$ lies above the graph of $y = \sin x \forall x \in \left(0, \frac{\pi}{2}\right)$.



Thus required area $\Delta = \int_0^{\pi/2}(x - \sin x)dx = \left(\frac{x^2}{2} + \cos x\right)_0^{\pi/2} = \frac{\pi^2 - 8}{8}$ sq. units.

Q.8 Answer:

$1/6$

Solution:

The points of intersection of curves are $x = 0$ and $x = 1$

$\therefore$ **required area** $= \int_0^1 (\sqrt{x} - x)dx = \left[\frac{2x^{3/2}}{3} - \frac{x^2}{2}\right]_0^1 = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$

Q.9 Answer:

A rea between the curve and the x-axis is less than $10\sqrt{5}$ sq. units

Solution:

$(x-2)^2 = -(y-5)$ **which is a parabola with vertex at** $(2, 5)$ **and inverted U type.**

This meets the x-axis at $(2 - \sqrt{5}, 0)$ **and** $(2 + \sqrt{5}, 0)$

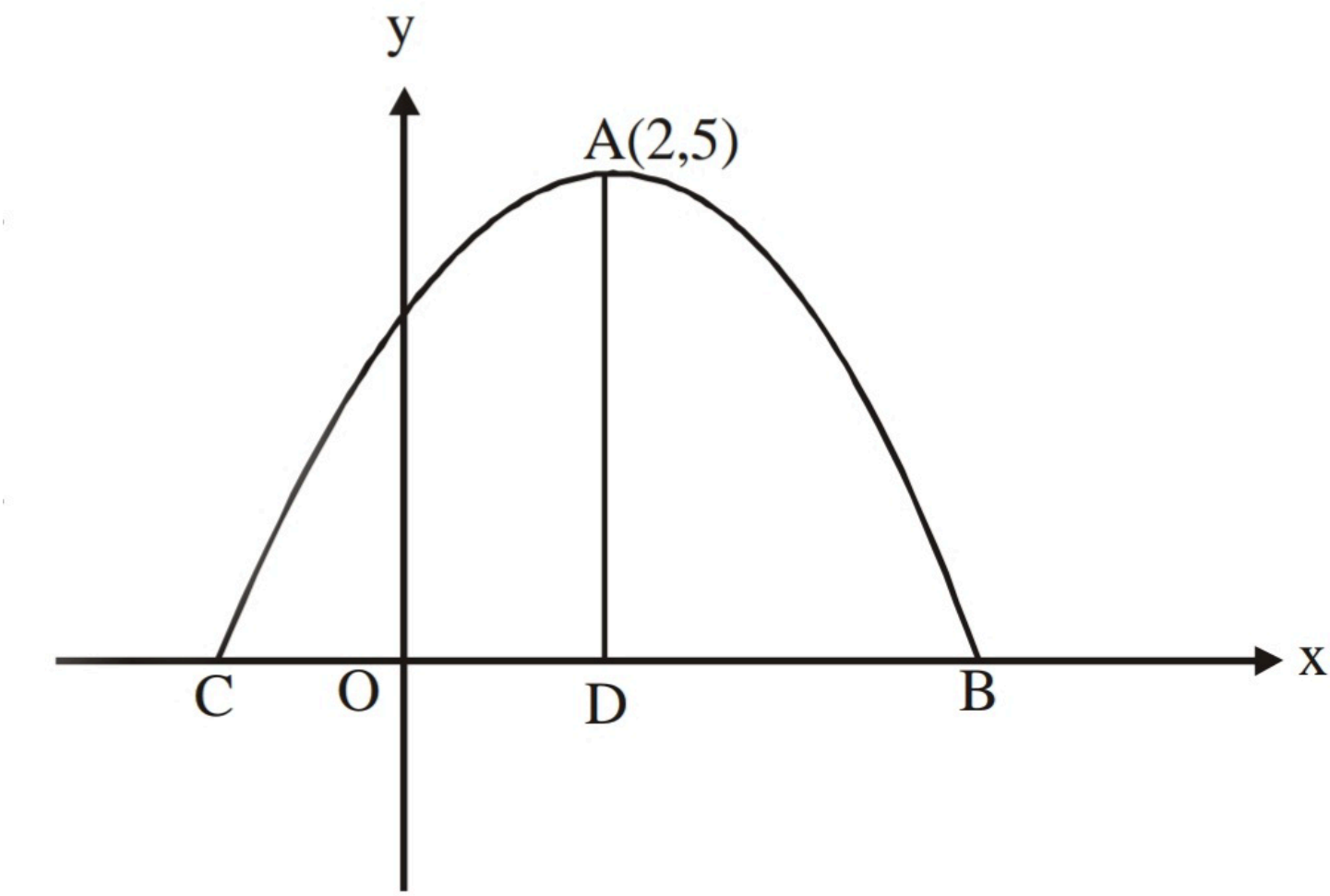
Area $<$ area of rectangle with CB and DA as sides $= (2\sqrt{5}) \times 5 = 10\sqrt{5}$

The line OA is $y = \frac{5}{2}x$

Area required $= \int_0^2 \left\{\left(1 + 4x - x^2\right) - \left(\frac{5}{2}x\right)\right\} dx = \left[x + \frac{3}{4}x^2 - \frac{x^3}{3}\right]_0^2 = \frac{7}{3}$

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$\int_0^{3/2} mxdx = \frac{1}{2}\int_0^{3/2} \left(1+4x-x^2\right) dx$



$\Rightarrow \frac{m}{2}\left(\frac{9}{4}\right) = \frac{1}{2}\left[\frac{3}{2}+2\left(\frac{9}{4}\right)-\frac{1}{3}\left(\frac{27}{8}\right)\right] \Rightarrow m = \frac{39}{18}$

$\therefore$ C **is not true**

D is evidently true.

Q.10 Answer:

-2

Solution:

Case I: When $m = 0$

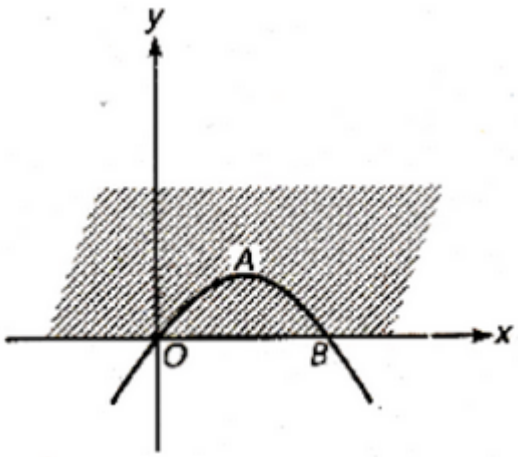
In this case $y = x - x^2$ **...(i)**

and $y = 0$ **...(ii)**

are two given curves, $y > 0$ **is total region above** x**-axis.**

Therefore, area between $y = x - x^2$ **and** $y = 0$ **is area between** $y = x - x^2$ **and above the** x**-axis**

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$$\therefore A = \int_0^1 \left(x - x^2\right) dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1$$
$$= \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \neq \frac{9}{2}$$

Hence, no solution exists.

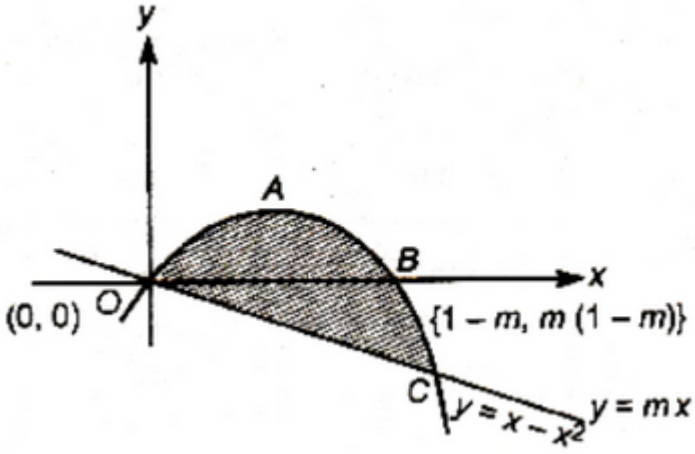
Case II: When $\mathrm{m} < 0$

In this case area between

$y = x - x^2$ **and** $y = mx$ **is**

Case II: When $m < 0$

In this case area between $y = x - x^2$ **and** $y = mx$ **is**



Put $\tan\frac{x}{2} = t \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

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$= \int_0^{\tan\frac{\pi}{8}} \frac{4tdt}{(1+t^2)\sqrt{1-t^2}}$

as $\tan\frac{\pi}{8} = \sqrt{2}-1$

So, $\int_0^{\sqrt{2}-1} \frac{4tdt}{(1+t^2)\sqrt{1-t^2}}$.

$OABCO$ **and points of intersection are** $(0,0)$ **and** $\{1-\mathrm{m}, \mathrm{m}(1-\mathrm{m})\}$

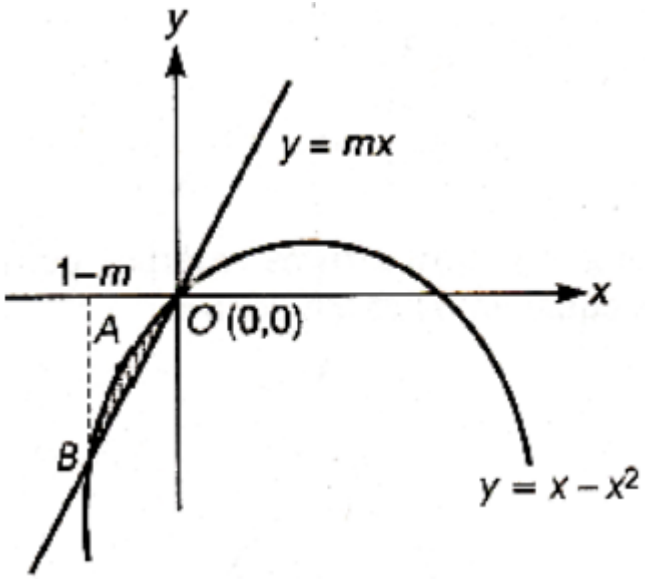
$\therefore$ **Area of curve** $OABCO$

$= \int_0^{1-m} \left[x - x^2 - mx\right] dx$

$$= \left[(1-m)\frac{x^2}{2} - \frac{x^3}{3}\right]_0^{1-m}$$
$$= \frac{1}{2}(1-m)^3 - \frac{1}{3}(1-m)^3 = \frac{1}{6}(1-m)^3$$
$$\therefore \quad \frac{1}{6}(1-m)^3 = \frac{9}{2} \quad (\text{given})$$
$$\Rightarrow \quad (1-m)^3 = 27$$
$$\Rightarrow \quad 1-m = 3$$
$$\Rightarrow \quad m = -2$$

Case III: When $m > 0$

In this case $y = mx$ **and** $y = x - x^2$ **intersect in** $(0,0)$ **and** $\{(1-m), m(1-m)\}$ **as shown in Fig.**



Area of shaded region

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$$= \int_{1-m}^{0} \left(x - x^2 - mx\right) dx$$

$$\left[= (1-m)\frac{x^2}{2} - \frac{x^3}{3}\right]_{1-m}$$

$$= -\frac{1}{2}(1-m)(1-m)^2 + \frac{1}{3}(1-m)^3$$

$$= -\frac{1}{6}(1-m)^3$$

$$\Rightarrow \frac{9}{2} = -\frac{1}{6}(1-m)^3 \text{ (given)}$$

$$\Rightarrow (1-m)^3 = -27$$

$$\Rightarrow (1-m) = -3$$

$$\Rightarrow m = 3 + 1 = 4.$$

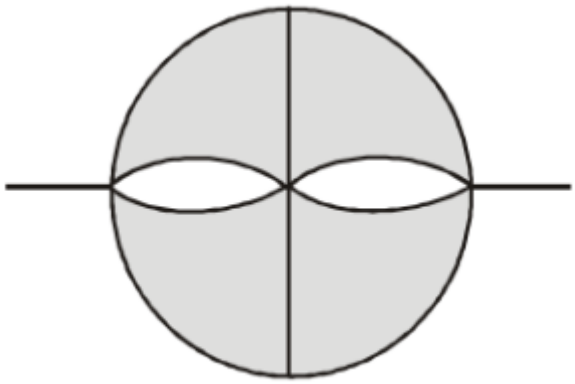
Therefore, (b) and (d) are the answers.

Q.11 Answer:

9

Solution:

Required area $= \pi\left(\pi^2\right) - 4\int_0^{\pi} \sin x dx = \pi^3 - 8$



Q.12 Answer:

$2\pi + \sqrt{3}$ sq. units

Solution:

(i) Given : $(x-y)f(x+y) - (x+y)f(x-y) = 4xy\left(x^2 - y^2\right)$

$= \left(x^2 - y^2\right)\left[(x+y)^2 - (x-y)^2\right]$

$= (x-y)(x+y)^3 - (x+y)(x-y)^3$

$\Rightarrow \mathrm{f(x+y)} = (\mathrm{x+y})^3 \Rightarrow \mathrm{f(x)} = \mathrm{x}^3, \mathrm{f(y)} = \mathrm{y}^3$

Now equation of given curves are

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$y^2 + x = 0$

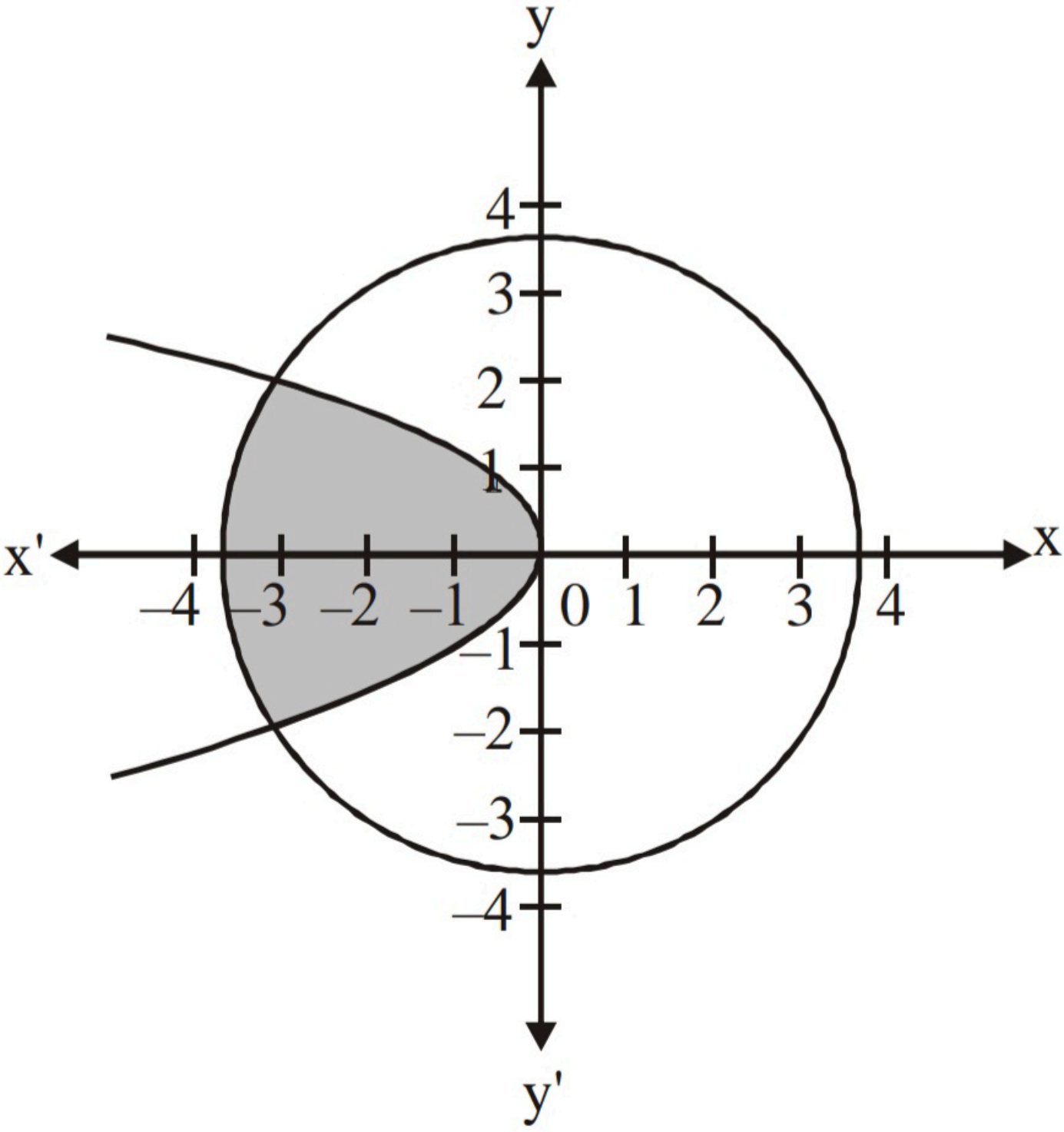
$x^2 + y^2 = 12$

$A = 2\left[\left|\int_{-2\sqrt{3}}^{-3} \sqrt{12 - x^2}dx\right| + \left|\int_{-3}^{0} \sqrt{-x}dx\right|\right]$

$I_1 = 2\int_{-2\sqrt{3}}^{-3} \sqrt{12 - x^2}dx = 2\int_{-\pi}^{-\pi}$

$= 12\left[\int_{-\pi/2}^{-\pi/3}(1 + \cos 2\theta)d\theta\right] = 12$

$= 12\left[\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right] = 2\pi - 3\sqrt{3}$

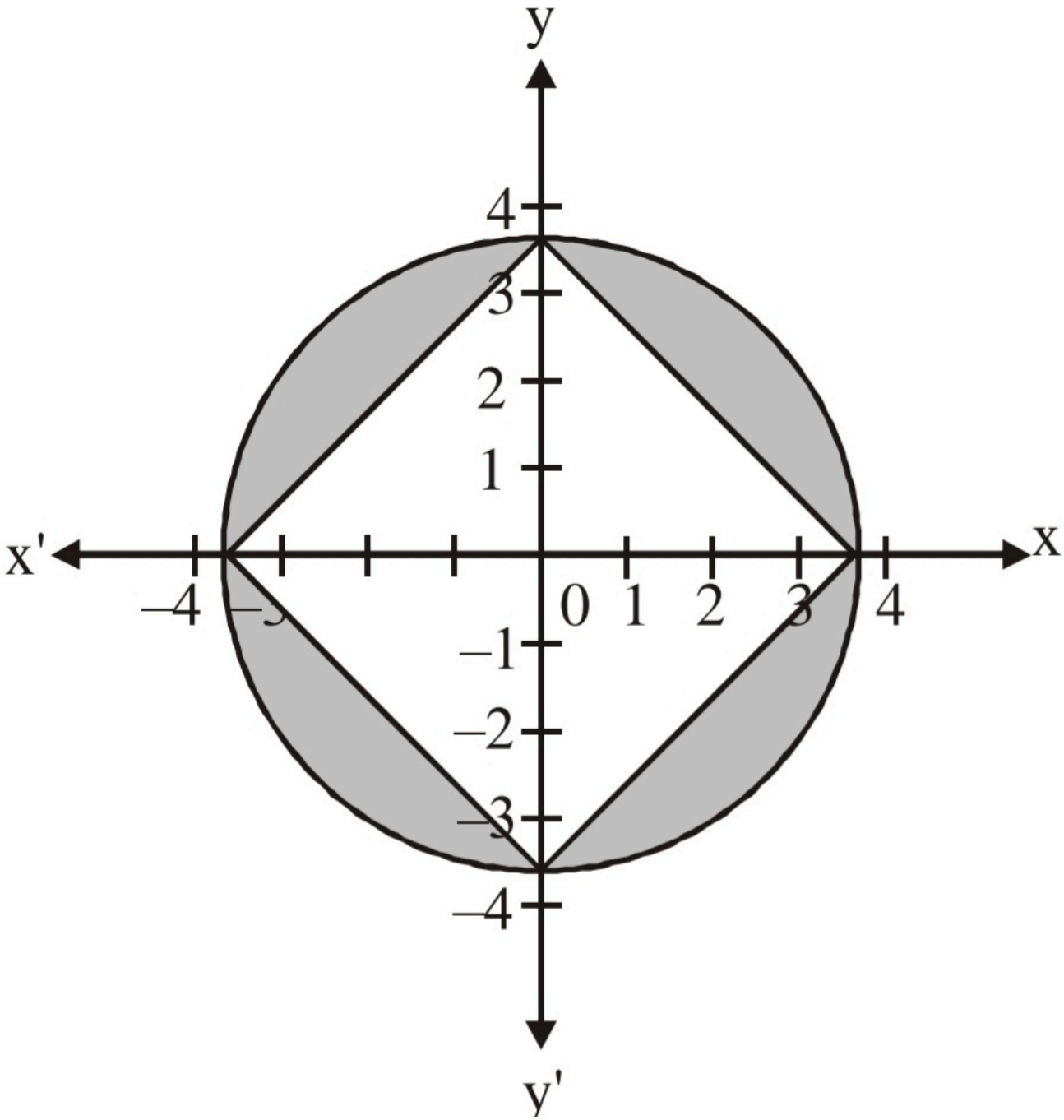


Solving eq. (1) and (2), we get $x = -3, y = \pm\sqrt{3}$

The area bounded by curves $I_2 = 2\int_{-3}^{0} \sqrt{-x}dx = \frac{2\left[(-x)^{3/2}\right]_{-3}^{0}}{-3/2} = -\frac{4}{3}\left[0 - 3^{3/2}\right] = 4\sqrt{3}$

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$A = 2\pi - 3\sqrt{3} + 4\sqrt{3} = 2\pi + \sqrt{3}$ **sq. units**



(ii) The required area is = area of circle − area of square

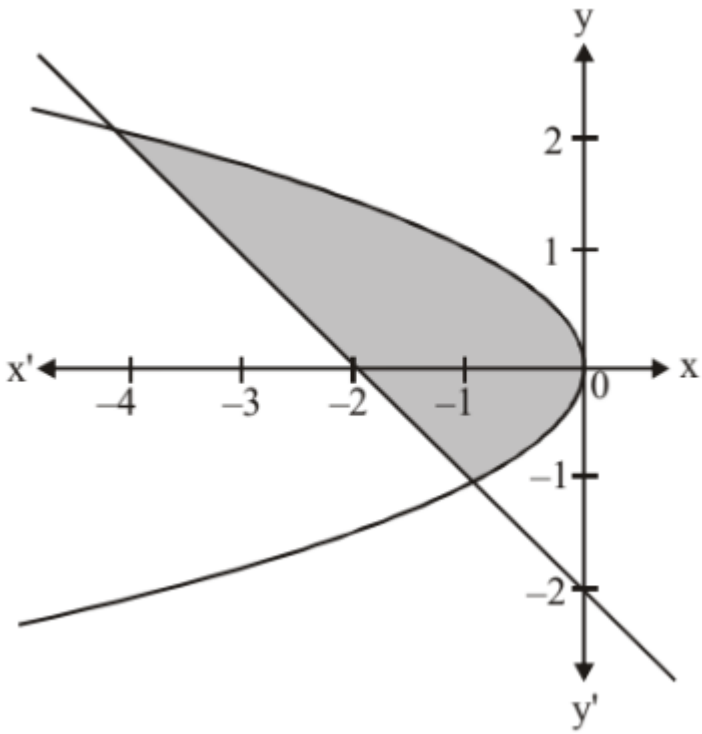
$= 12\pi - 24$ sq. units

(iii) The required area $= \int_{-1}^{2} \left(-y^2 - (-y - 2)dy\right.$

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$$= \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2}$$

$$= \left[\frac{4}{2} + 4 - \frac{8}{3} - \left(\frac{1}{2} - 2 + \frac{1}{3}\right)\right] = \frac{9}{2} \text{ sq. units}$$



Q.13 Answer:

$12\pi - 24$ sq. units

Solution:

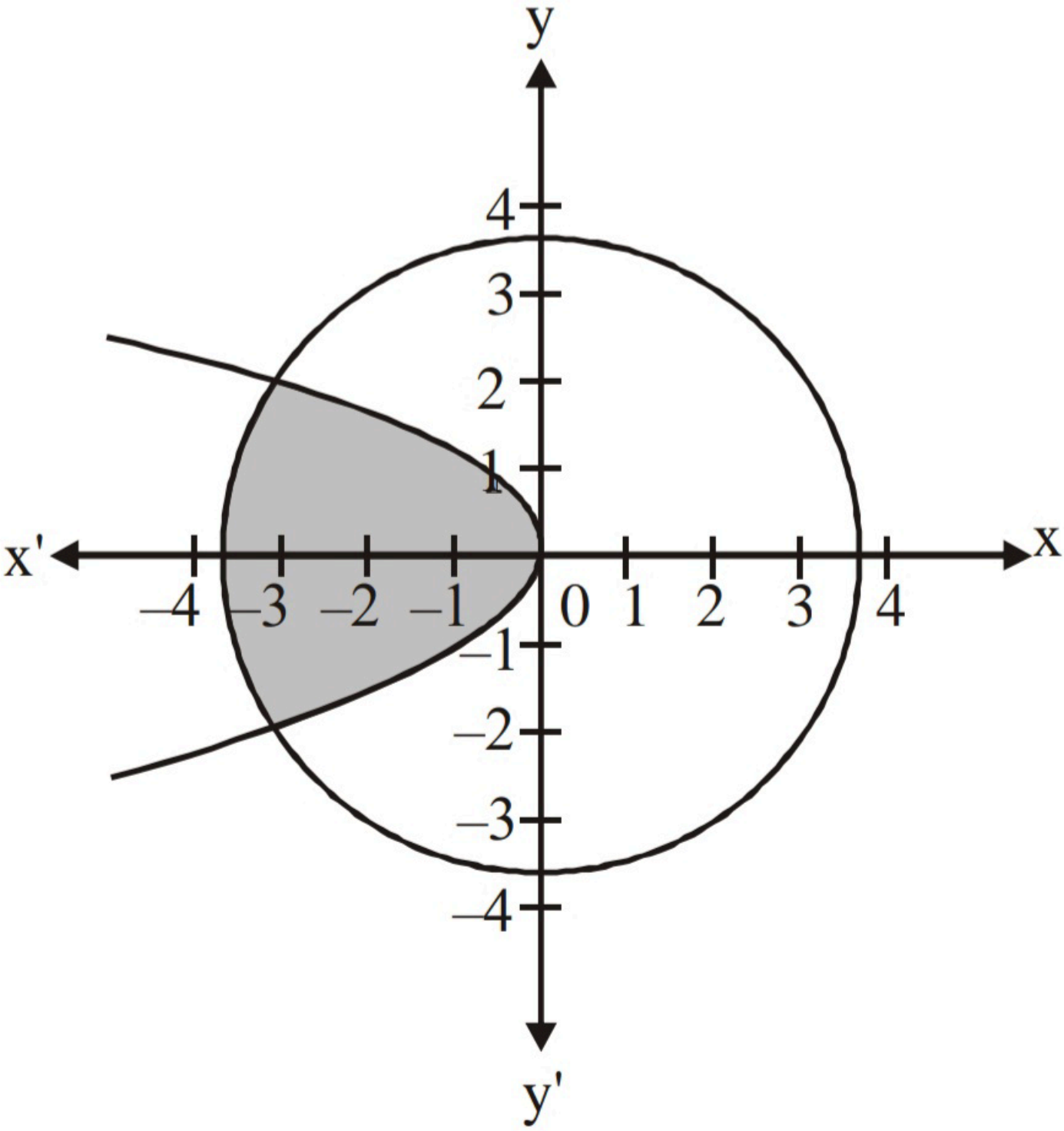
(i) Given : $(x-y)f(x+y) - (x+y)f(x-y) = 4xy\left(x^2-y^2\right) =$
$\left(x^2-y^2\right)\left[(x+y)^2-(x-y)^2\right]$
$= (x-y)(x+y)^3 - (x+y)(x-y)^3$
$\Rightarrow \mathrm{f(x+y)} = \mathrm{(x+y)^3} \Rightarrow \mathrm{f(x)} = \mathrm{x^3}, \mathrm{f(y)} = \mathrm{y^3}$

Now equation of given curves are

$\mathrm{y^2 + x = 0}$

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$x^2 + y^2 = 12$ ..... (2)



Solving eq. (1) and (2), we get $x = -3, y = \pm\sqrt{3}$

The area bounded by curves

$$A = 2\left[\left|\int_{-2\sqrt{3}}^{-3}\sqrt{12 - x^2}dx\right| + \left|\int_{-3}^{0}\sqrt{-x}dx\right|\right]$$

$$I_1 = 2\int_{-2\sqrt{3}}^{-3}\sqrt{12 - x^2}dx = 2\int_{-\pi/2}^{-\pi/3} 12\cos^2\theta d\theta$$

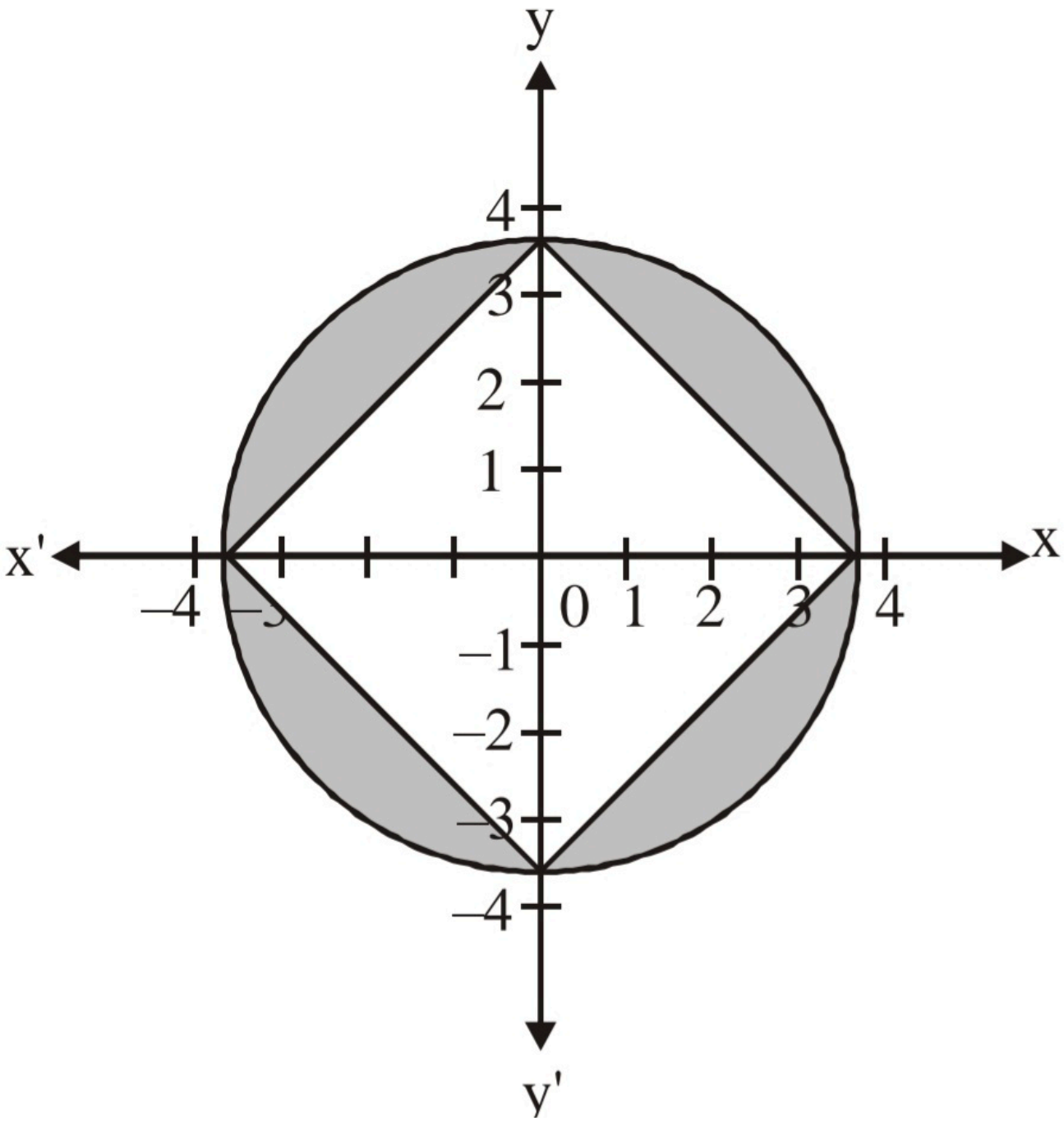
$$= 12\left[\int_{-\pi/2}^{-\pi/3}(1 + \cos 2\theta)d\theta\right] = 12\left[\theta + \frac{\sin\theta}{2}\right]_{-\pi/2}^{-\pi/3} = 12\left[-\frac{\pi}{3} - \frac{\sqrt{3}}{4} + \frac{\pi}{2}\right]$$

$$= 12\left[\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right] = 2\pi - 3\sqrt{3}$$

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$I_2 = 2\int_{-3}^{0}\sqrt{-x}dx = \frac{2\left[(-x)^{3/2}\right]_{-3}^{0}}{-3/2} = -\frac{4}{3}\left[0 - 3^{3/2}\right] = 4\sqrt{3}$

$A = 2\pi - 3\sqrt{3} + 4\sqrt{3} = 2\pi + \sqrt{3}$ sq. units



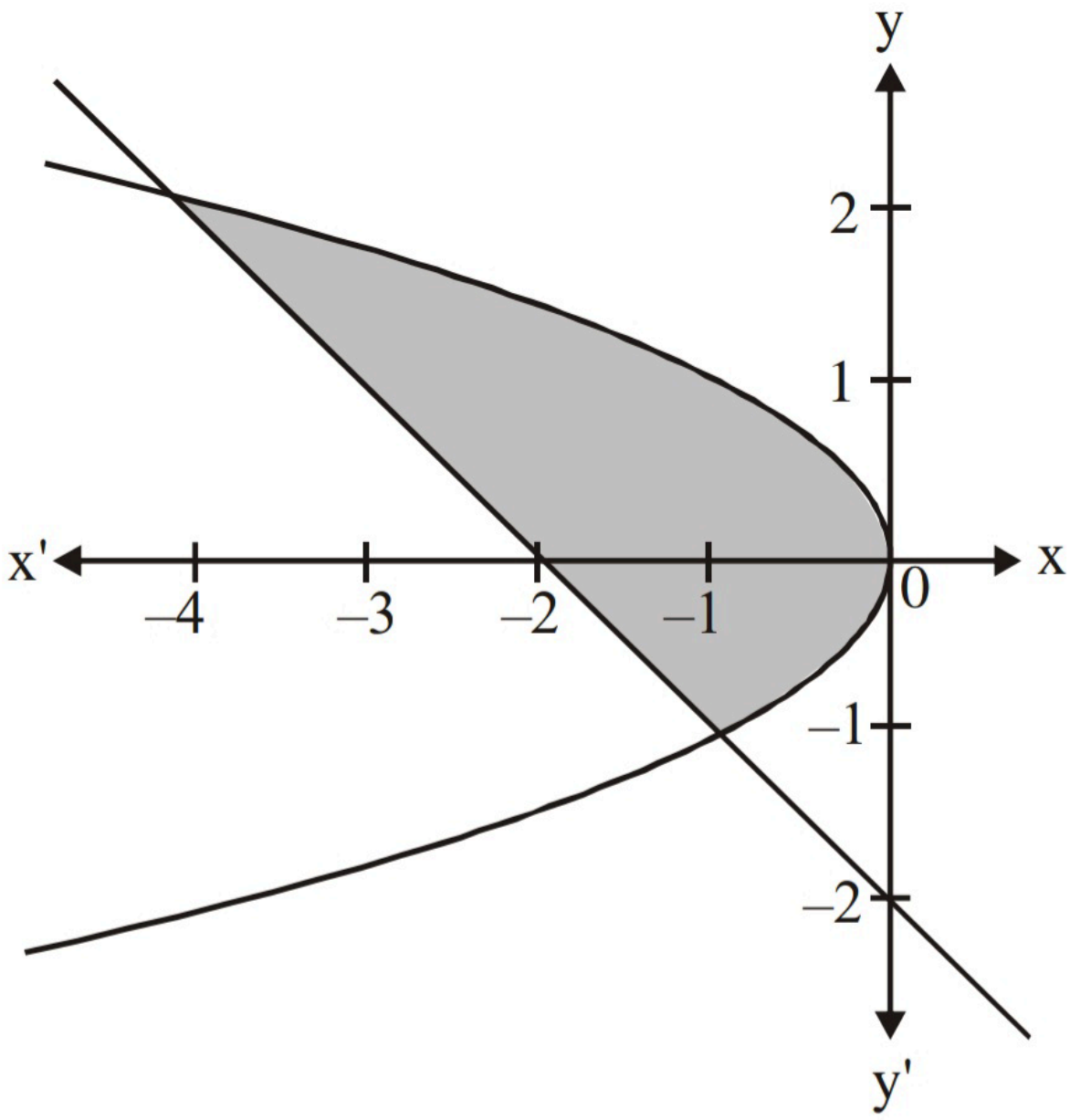
(ii) The required area is = area of circle − area of square = $12\pi - 24$ sq. units

(iii) The required area = $\int_{-1}^{2}\left(-y^2 - (-y-2)dy\right.$

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$= \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2}$

$= \left[\frac{4}{2} + 4 - \frac{8}{3} - \left(\frac{1}{2} - 2 + \frac{1}{3}\right)\right] = \frac{9}{2}$ sq. units



Q.14 Answer:

$9/2$ sq. units

Solution:

(i) Given : $(x-y)f(x+y) - (x+y)f(x-y) = 4xy\left(x^2-y^2\right) =$
$\left(x^2-y^2\right)\left[(x+y)^2-(x-y)^2\right]$

$= (x-y)(x+y)^3 - (x+y)(x-y)^3$

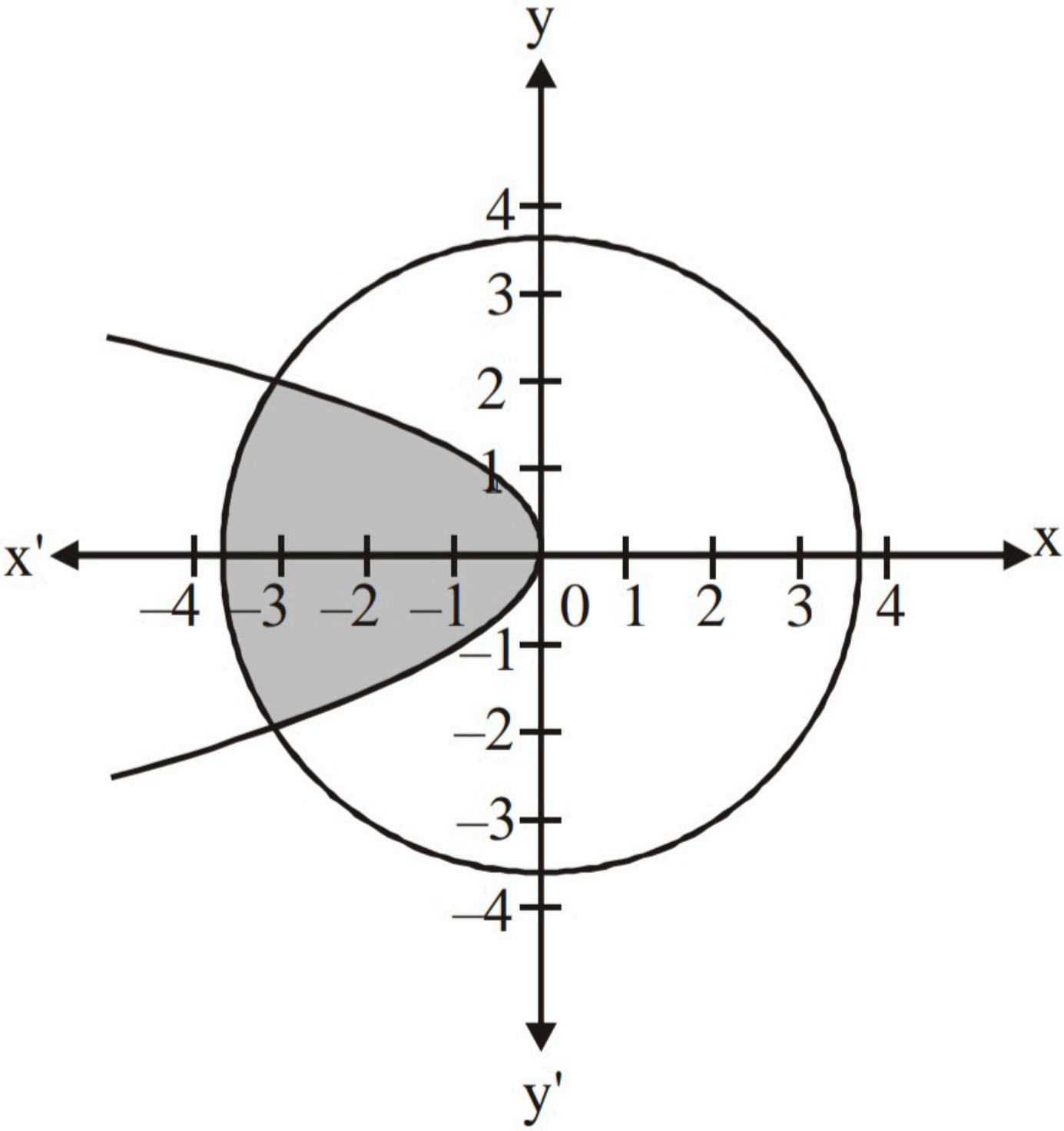
$\Rightarrow \mathrm{f}(\mathrm{x}+\mathrm{y}) = (\mathrm{x}+\mathrm{y})^3 \Rightarrow \mathrm{f}(\mathrm{x}) = \mathrm{x}^3, \mathrm{f}(\mathrm{y}) = \mathrm{y}^3$

Now equation of given curves are

$\mathrm{y}^2 + \mathrm{x} = 0$

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$x^2 + y^2 = 12$ ..... (2)



Solving eq. (1) and (2), we get $x = -3, y = \pm\sqrt{3}$

The area bounded by curves

$$A = 2\left[\left|\int_{-2\sqrt{3}}^{-3}\sqrt{12 - x^2}dx\right| + \left|\int_{-3}^{0}\sqrt{-x}dx\right|\right]$$

$$I_1 = 2\int_{-2\sqrt{3}}^{-3}\sqrt{12 - x^2}dx = 2\int_{-\pi/2}^{-\pi/3} 12\cos^2\theta d\theta$$

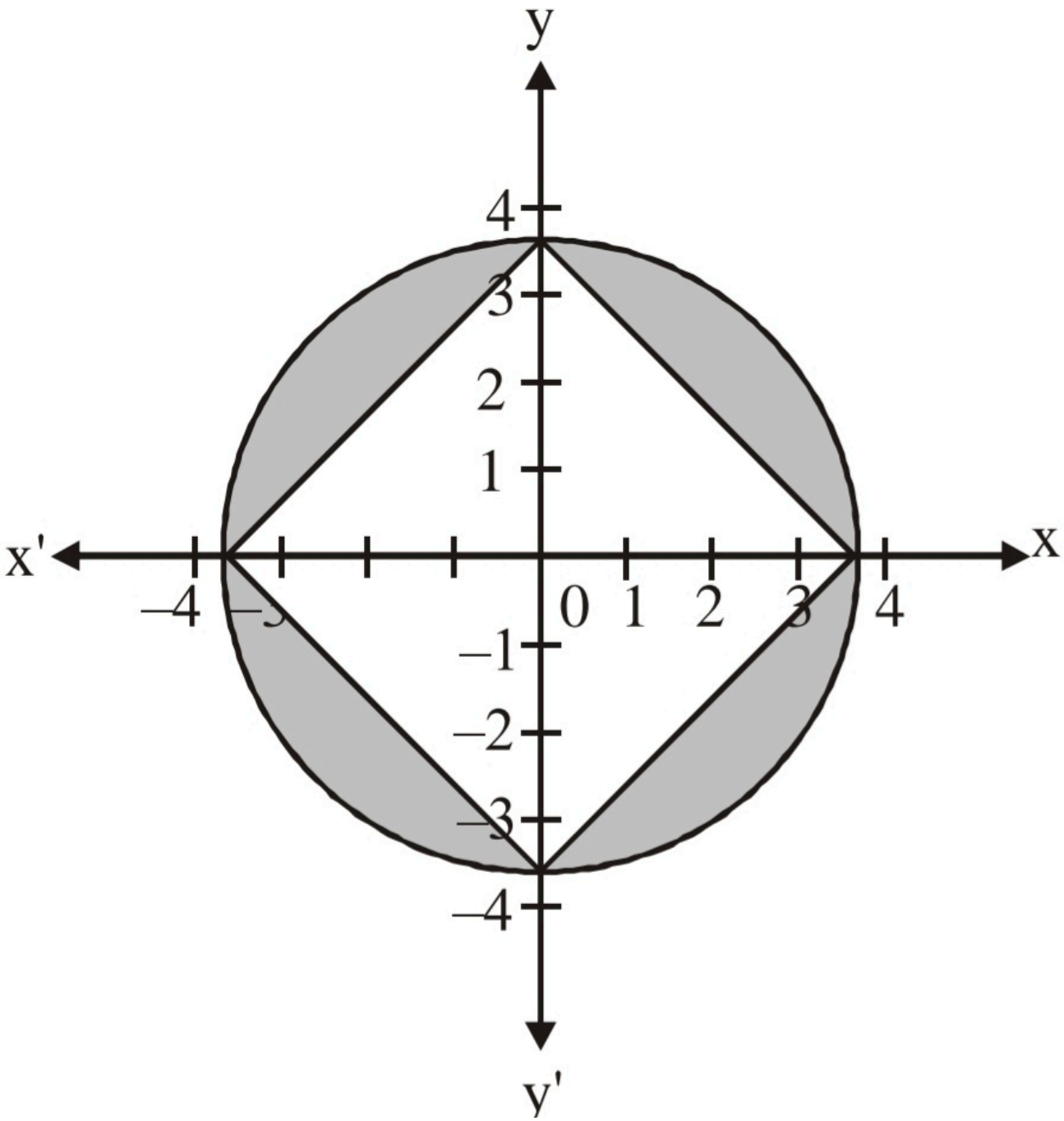
$$= 12\left[\int_{-\pi/2}^{-\pi/3}(1 + \cos 2\theta)d\theta\right] = 12\left[\theta + \frac{\sin\theta}{2}\right]_{-\pi/2}^{-\pi/3} = 12\left[-\frac{\pi}{3} - \frac{\sqrt{3}}{4} + \frac{\pi}{2}\right]$$

$$= 12\left[\frac{\pi}{6} - \frac{\sqrt{3}}{4}\right] = 2\pi - 3\sqrt{3}$$

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$I_2 = 2\int_{-3}^{0}\sqrt{-x}dx = \frac{2\left[(-x)^{3/2}\right]_{-3}^{0}}{-3/2} = -\frac{4}{3}\left[0 - 3^{3/2}\right] = 4\sqrt{3}$

$A = 2\pi - 3\sqrt{3} + 4\sqrt{3} = 2\pi + \sqrt{3}$ sq. units



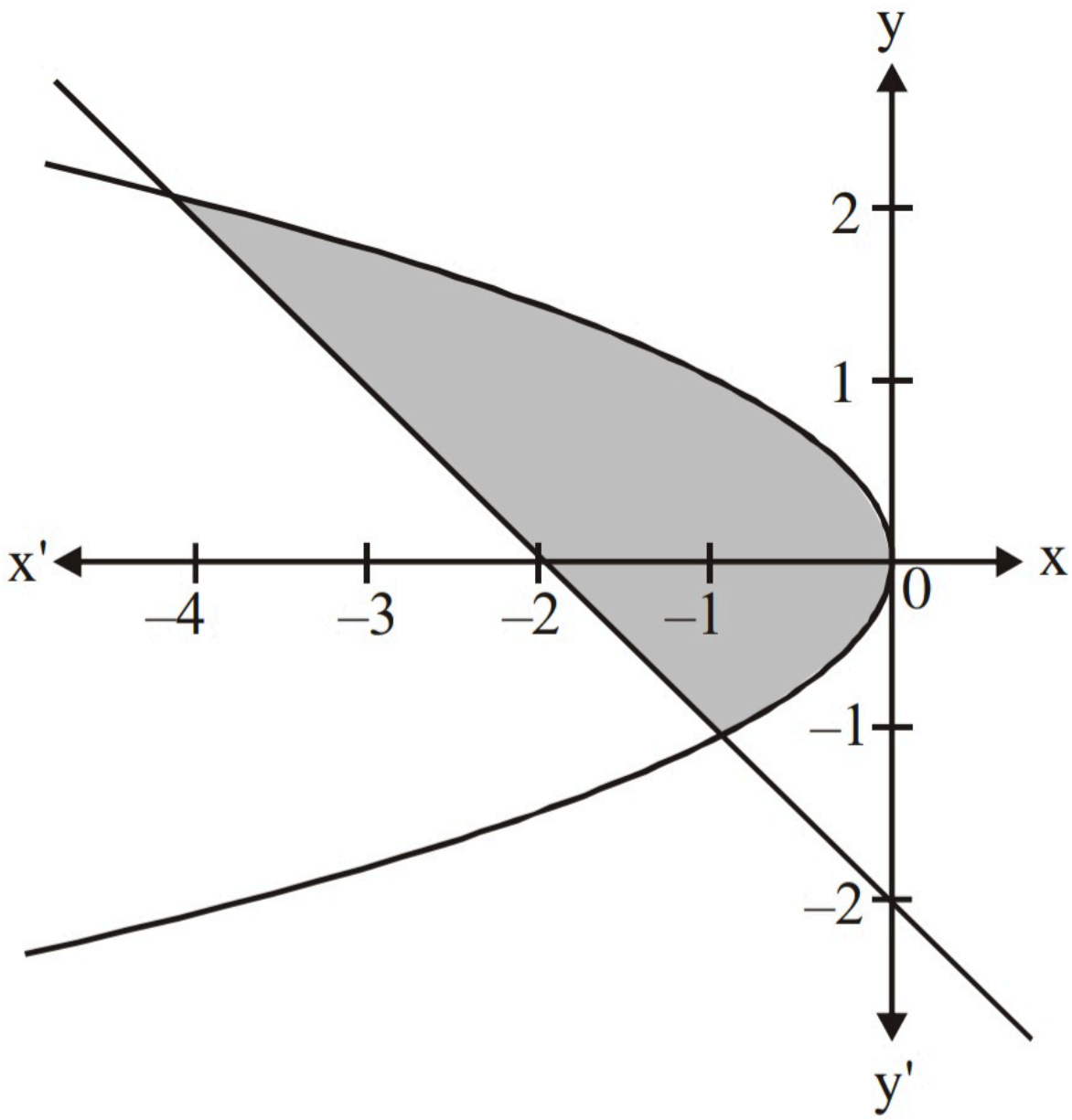
(ii) The required area is = area of circle − area of square = $12\pi - 24$ sq. units

(iii) The required area = $\int_{-1}^{2}\left(-y^2 - (-y-2)dy\right.$

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$= \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2}$

$= \left[\frac{4}{2} + 4 - \frac{8}{3} - \left(\frac{1}{2} - 2 + \frac{1}{3}\right)\right] = \frac{9}{2}$ sq. units



Q.15 Answer:

$\frac{\pi}{2} + 1$ sq. units

Solution:

(i) For $-1 \leq x < 0$

$$\left(y - e^{\sin^{-1} x}\right)^2 = 2 - x^2$$

$$y = e^{\sin^{-1} x} \pm \sqrt{2 - x^2}$$

$$A = \int_{-1}^{0} \left(e^{\sin^{-1} x} + \sqrt{2 - x^2}\right) dx - \left(e^{\sin^{-1} x} - \sqrt{2 - x^2}\right) dx = 2\int_{-1}^{0} \sqrt{2 - x^2} dx$$

$$= 2\left(\frac{1}{2} x\sqrt{2 - x^2}\Big|_{-1}^{0} + \frac{2}{2}\sin^{-1}\frac{x}{\sqrt{2}}\Big|_{-1}^{0}\right) = \left[1 + 2\left(0 - \left(-\frac{\pi}{4}\right)\right)\right] = \frac{\pi}{2} + 1 \text{ sq. units}$$

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For $0 \leq x < 1, y = \sin^{-1} x \pm \sqrt{1-x^2}$

$$A = 2\int_0^1 \sqrt{1-x^2}dx = 2\left[\frac{x}{2}\sqrt{1-x^2}\Big|_0^1 + \frac{1}{2}\sin^{-1}\frac{x}{1}\Big|_0^1\right] = 0 + \sin^{-1}(1) = \frac{\pi}{2} \text{ sq. units}$$

Total area $= \left(\frac{\pi}{2}+1\right) + \frac{\pi}{2} = \pi + 1$

(ii) Ratio $= \frac{\frac{\pi}{2}+1}{\frac{\pi}{2}} = \frac{\pi+2}{\pi}$ **(iii)** $A = 2\int_0^{1/2}\sqrt{1-x^2}dx = 2\left[\frac{x}{2}\sqrt{1-x^2}\Big|_0^{1/2} - \frac{1}{2}\sin^{-1}x\Big|_0^{1/2}\right] = \frac{\sqrt{3}}{4} + \frac{\pi}{6}$ **sq. units**

Q.16 Answer:

$\pi + 2 : \pi$

Solution:

(i) For $-1 \leq x < 0$

$$\left(y - e^{\sin^{-1}x}\right)^2 = 2 - x^2$$

$$y = e^{\sin^{-1}x} \pm \sqrt{2-x^2}$$

$$A = \int_{-1}^0 \left(e^{\sin^{-1}x} + \sqrt{2-x^2}\right)dx - \left(e^{\sin^{-1}x} - \sqrt{2-x^2}\right)dx = 2\int_{-1}^0 \sqrt{2-x^2}dx$$

$$= 2\left(\frac{1}{2}x\sqrt{2-x^2}\Big|_{-1}^0 + \frac{2}{2}\sin^{-1}\frac{x}{\sqrt{2}}\Big|_{-1}^0\right) = \left[1 + 2\left(0 - \left(-\frac{\pi}{4}\right)\right)\right] = \frac{\pi}{2} + 1 \text{ sq. units}$$

For $0 \leq x < 1, y = \sin^{-1} x \pm \sqrt{1-x^2}$

$\mathrm{A} = 2\int_0^1 \sqrt{1-\mathrm{x}^2}\,\mathrm{dx} = 2\left[\frac{\mathrm{x}}{2}\sqrt{1-\mathrm{x}^2}\Big|_0^1 + \frac{1}{2}\sin^{-1}\frac{\mathrm{x}}{1}\Big|_0^1\right] = 0 + \sin^{-1}(1) = \frac{\pi}{2}$ **sq. units**

Total area $= \left(\frac{\pi}{2}+1\right) + \frac{\pi}{2} = \pi + 1$

(ii) Ratio $= \frac{\frac{\pi}{2}+1}{\frac{\pi}{2}} = \frac{\pi+2}{\pi}$

(iii) $\mathrm{A} = 2\int_0^{1/2}\sqrt{1-\mathrm{x}^2}\,\mathrm{dx} = 2\left[\frac{\mathrm{x}}{2}\sqrt{1-\mathrm{x}^2}\Big|_0^{1/2} - \frac{1}{2}\sin^{-1}\mathrm{x}\Big|_0^{1/2}\right] = \frac{\sqrt{3}}{4} + \frac{\pi}{6}$ **sq. units**

Q.17 Answer:

$\frac{\sqrt{3}}{4} + \frac{\pi}{6}$ sq. units

Solution:

(i) For $-1 \leq x < 0$

$$\left(y - e^{\sin^{-1}x}\right)^2 = 2 - x^2$$

$$y = e^{\sin^{-1}x} \pm \sqrt{2-x^2}$$

$$A = \int_{-1}^0 \left(e^{\sin^{-1}x} + \sqrt{2-x^2}\right)dx - \left(e^{\sin^{-1}x} - \sqrt{2-x^2}\right)dx = 2\int_{-1}^0 \sqrt{2-x^2}dx$$

$$= 2\left(\frac{1}{2}x\sqrt{2-x^2}\Big|_{-1}^0 + \frac{2}{2}\sin^{-1}\frac{x}{\sqrt{2}}\Big|_{-1}^0\right) = \left[1 + 2\left(0 - \left(-\frac{\pi}{4}\right)\right)\right] = \frac{\pi}{2} + 1 \text{ sq. units}$$

For $0 \leq x < 1, y = \sin^{-1} x \pm \sqrt{1-x^2}$

$\mathrm{A} = 2\int_0^1 \sqrt{1-\mathrm{x}^2}\,\mathrm{dx} = 2\left[\frac{\mathrm{x}}{2}\sqrt{1-\mathrm{x}^2}\Big|_0^1 + \frac{1}{2}\sin^{-1}\frac{\mathrm{x}}{1}\Big|_0^1\right] = 0 + \sin^{-1}(1) = \frac{\pi}{2}$ **sq. units**

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Total area $= \left(\frac{\pi}{2}+1\right)+\frac{\pi}{2}=\pi+1$

(ii) Ratio $= \frac{\frac{\pi}{2}+1}{\frac{\pi}{2}} = \frac{\pi+2}{\pi}$

$$\text{(iii) } A = 2\int_0^{1/2}\sqrt{1-x^2}\,dx = 2\left[\frac{x}{2}\sqrt{1-x^2}\Big|_0^{1/2} - \frac{1}{2}\sin^{-1}x\Big|_0^{1/2}\right] = \frac{\sqrt{3}}{4}+\frac{\pi}{6} \text{ sq. units}$$

Q.18 Answer:

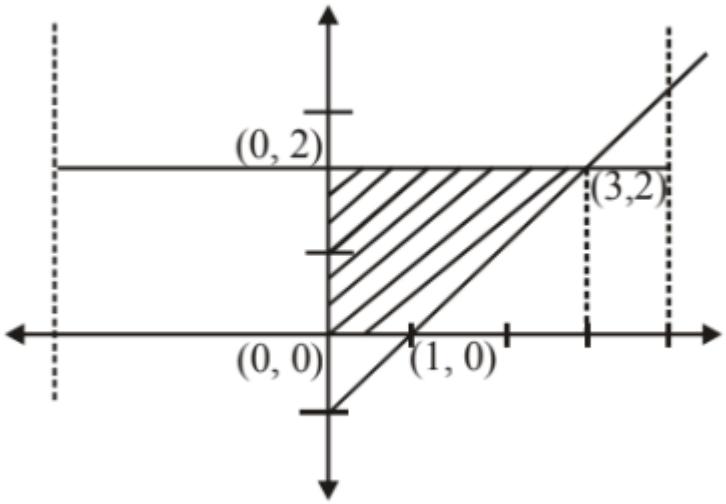
A-p ; B-q ; C-r ; D-q

Solution:

(A) – p, (B) – q, (C) – r, (D) – q

(A) $x \in (-8,8), y = 2\; x \in (-8\sqrt{2},-8]\cup[8,8\sqrt{2}), \quad y = 3$

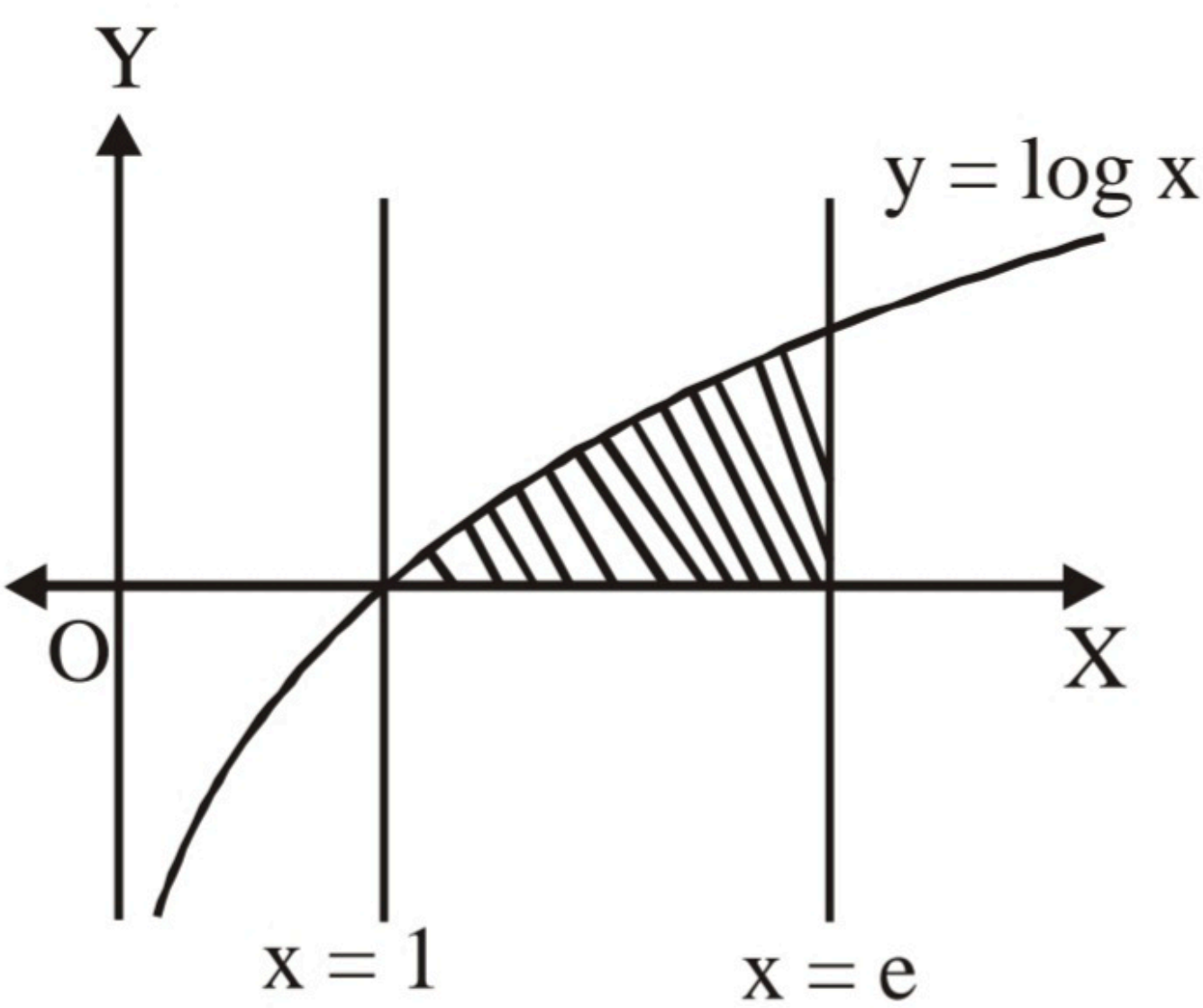
Required area $= 2\times 3 - \frac{1}{2}\times 2\times 2 = 6-2 = 4$ **sq. units**



(B) The required area $= \int_1^e y\,dx = \int_1^e \log x\,dx$

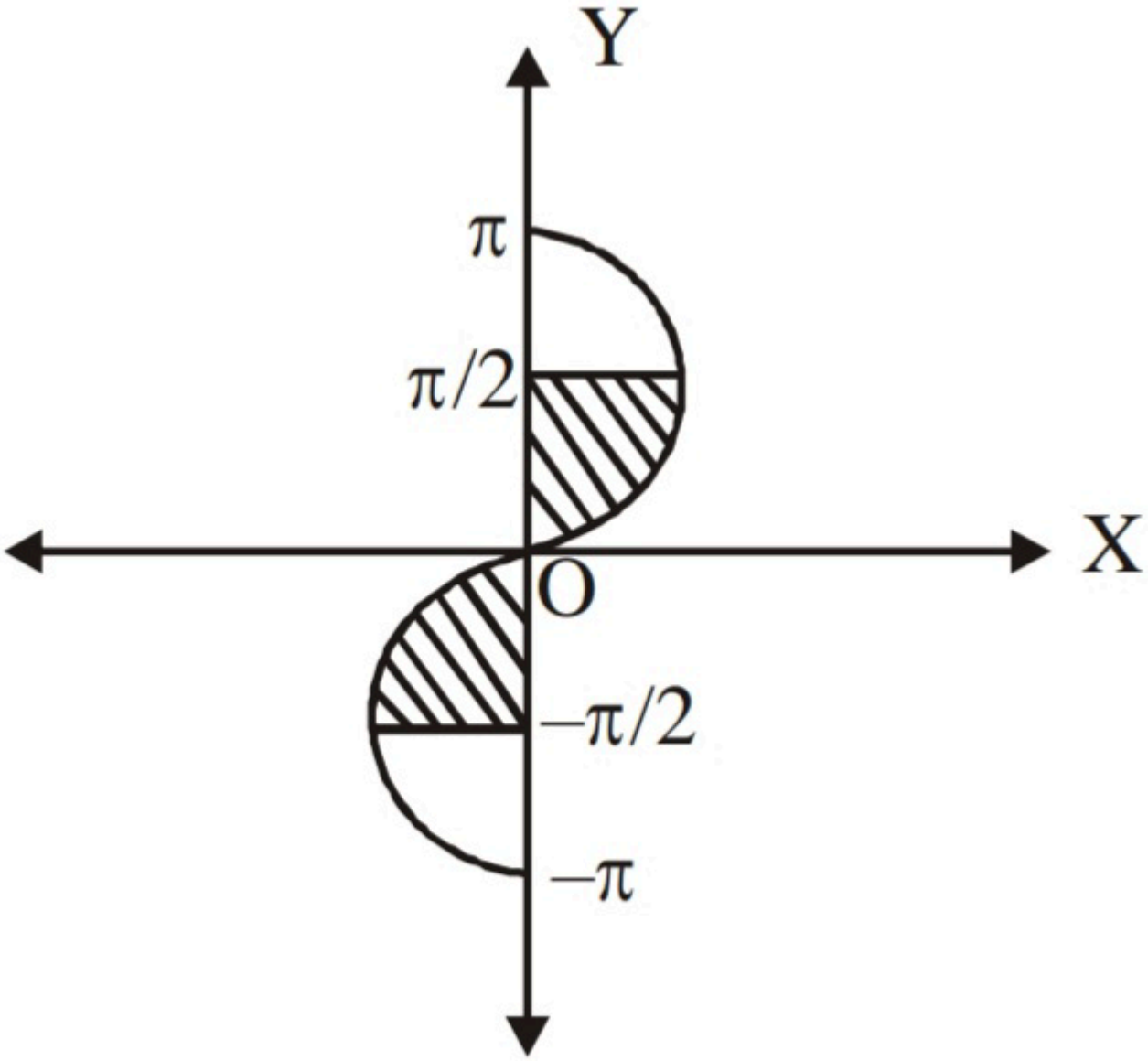
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$[x \log x - x]_1^e = (e \log e - e) - (0 - 1) = e - e + 1 = 1$



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(C)



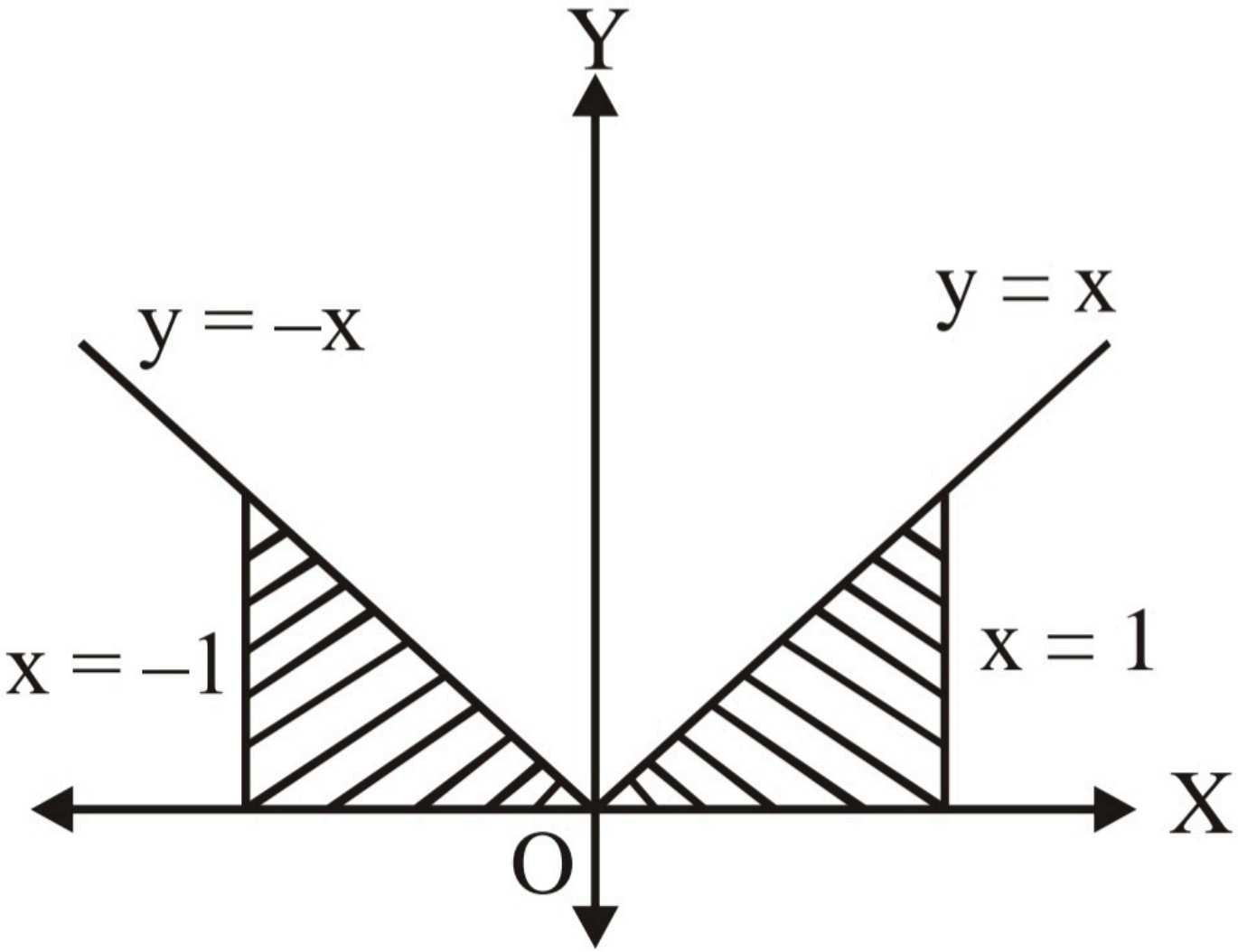
The required area,

$=2\int_0^{\pi/2} x\,dy$ where $y = \sin^{-1} x$ i.e. x

$= 2\int_0^{\pi/2} \sin y\,dy = 2\left[\cos y\left[0^{\pi/2} = 2\right.\right.$

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(D) The required area $= 2\int_0^1 \mathrm{ydx}$, **where** $\mathrm{y} = \mathrm{x} = 2\int_0^1 \mathrm{xdx} = 2\left[\frac{\mathrm{x}^2}{2}\right]_0^1 = 1$



Q.19 Answer:

—

Solution:

$y = a - bx^2$

when $\mathrm{x} = 2, \mathrm{y} = 1$

$1 = \mathrm{a} - 4\,\mathrm{b} \Rightarrow \mathrm{a} = 1 + 4\,\mathrm{b}$

$A = 2\int_0^{\sqrt{a/b}} \left(a - bx^2\right) dx = 2\left[ax - \frac{bx^3}{3}\right]_0^{\sqrt{a/b}} = 2\left[\frac{a\sqrt{a}}{\sqrt{b}} - \frac{b}{3}\frac{a\sqrt{a}}{b\sqrt{b}}\right]$

$= \frac{2}{3}\left[\frac{3\mathrm{a}\sqrt{\mathrm{a}}}{\sqrt{\mathrm{b}}} - \frac{\mathrm{a}\sqrt{\mathrm{a}}}{\sqrt{\mathrm{b}}}\right] = \frac{2}{3}\left[\frac{2\mathrm{a}\sqrt{\mathrm{a}}}{\sqrt{\mathrm{b}}}\right] = \frac{4}{3}\frac{(1+4\,\mathrm{b})\sqrt{1+4\,\mathrm{b}}}{\sqrt{\mathrm{b}}}$

Since area (A) **has to be minimum**

$\frac{dA}{db} = 0.$

As a result we get,

$\Rightarrow 12\,\mathrm{b} = 1 + 4\,\mathrm{b} \Rightarrow 8\,\mathrm{b} = 1 \Rightarrow \mathrm{b} = \frac{1}{8}, \mathrm{a} = \frac{3}{2} \therefore \quad A_{\min} = \frac{4}{3}\left(\frac{3}{2}\right)\sqrt{\frac{3}{2}} \cdot 2\sqrt{2} = \frac{4\sqrt{3}\sqrt{2}}{\sqrt{2}}$

$\therefore$ **Minimum area** $= \sqrt{48} \Rightarrow \mathrm{a} = 48 \Rightarrow \frac{\mathrm{A}}{8} = 6$

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Q.20 Answer:

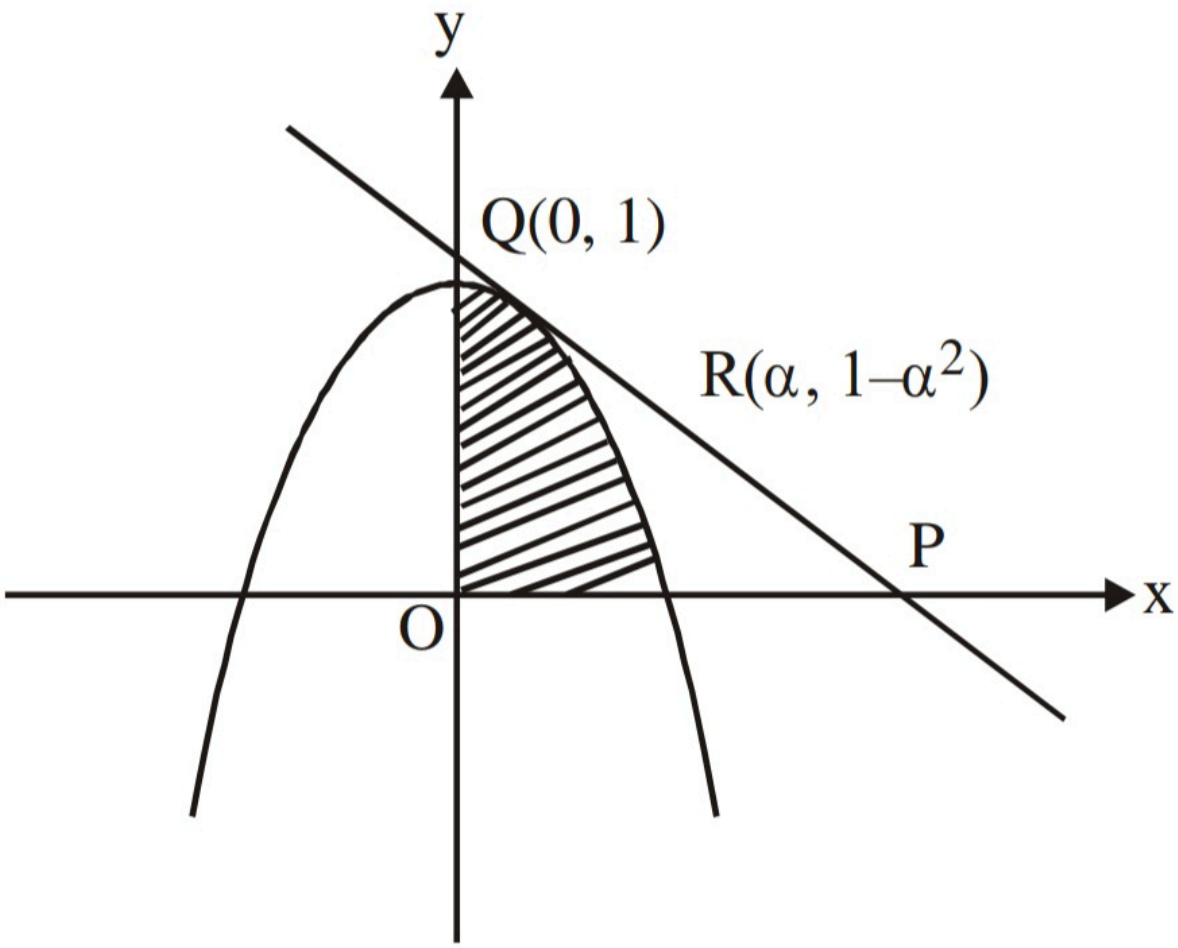
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Solution:

A_1 = **area under the curve and axes**

$= \int_0^1 (1-x^2)\,dx = \left[x - \frac{x^3}{3}\right]_0^1 = \frac{2}{3} \ldots\ldots\ldots\ldots (1)$

Now, $y = 1 - x^2 \quad \therefore y' = -2x \quad \therefore y'(\alpha, 1-\alpha^2) = -2\alpha$ **Equation of tangent to the curve** $y = 1 - x^2$



$\Rightarrow \quad (y - (1-\alpha^2)) = -2\alpha(x - \alpha)$

$\Rightarrow 2\alpha x + y = \alpha^2 + 1$

$\therefore \quad P \equiv \left(\frac{\alpha^2+1}{2\alpha}, 0\right); Q \equiv (0, \alpha^2+1)$

$A =$ **Area of triangle** $POQ = \frac{1}{2}(OP)(OQ) = \frac{1}{4}\frac{(\alpha^2+1)^2}{\alpha}$

$\therefore \quad \frac{dA}{d\alpha} = \frac{1}{4}\frac{\alpha(\alpha^2+1)2\alpha - (\alpha^2+1)^2}{\alpha^2} = \frac{1}{4}\frac{3\alpha^4+2\alpha^2-1}{\alpha^2}$

For maximum/minimum, $\frac{dA}{d\alpha} = 0 \Rightarrow 3\alpha^4 + 2\alpha^2 - 1 = 0 \Rightarrow \alpha^2 = \frac{1}{3} \Rightarrow \alpha = \pm\frac{1}{\sqrt{3}}$

Since, $0 < \alpha < 1 \Rightarrow \alpha = \frac{1}{\sqrt{3}}, \frac{d^2A}{d\alpha^2} > 0$

Hence, A **is minimum,** $A = \frac{1}{4}\frac{\left(\frac{1}{3}+1\right)^2}{\frac{1}{\sqrt{3}}} = \frac{4}{3\sqrt{3}}$

Since $A = kA_1$ **(given)** $\Rightarrow \frac{4}{3\sqrt{3}} = \frac{2}{k} \Rightarrow k = \frac{2}{\sqrt{3}}$

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